Literature Survey

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Several research studies have explored the application of Machine Learning in agriculture to improve crop selection and farming productivity. Crop recommendation systems analyze soil nutrients and environmental conditions to identify the most suitable crop for cultivation. Commonly used input parameters include Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH, and rainfall.

Researchers have implemented various Machine Learning algorithms such as **Logistic Regression**, **Decision Trees**, **Random Forests**, **K-Nearest Neighbors (KNN)**, and **K-Means Clustering** to predict suitable crops based on agricultural data. These techniques have demonstrated promising results in improving prediction accuracy and supporting data-driven farming decisions.

The literature also highlights the importance of **data preprocessing**, **exploratory data analysis (EDA)**, **feature scaling**, and **model evaluation** in developing reliable crop recommendation systems. Proper analysis of agricultural datasets helps identify relationships among soil nutrients and climatic factors, resulting in more accurate and efficient prediction models.

Although many existing crop recommendation systems provide useful predictions, several are limited in terms of usability, scalability, or model comparison. Some systems focus on a single algorithm without evaluating multiple approaches, while others lack an easy-to-use web interface for farmers.

The proposed **OptiCrop: Smart Agricultural Production Optimization Engine** addresses these limitations by implementing a structured Machine Learning workflow that includes dataset collection, exploratory data analysis, data preprocessing, K-Means clustering, Logistic Regression-based crop recommendation, model evaluation, and deployment through a simple web application. The system aims to provide accurate crop recommendations that help farmers improve productivity, optimize resource utilization, and support informed agricultural decision-making.